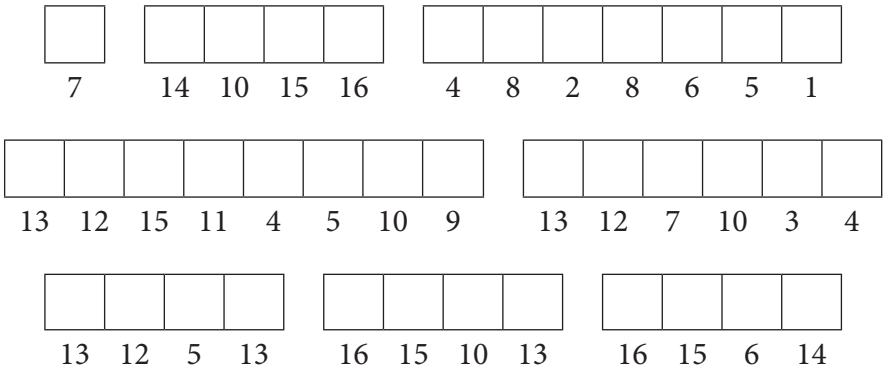
Name: _____

WORKSHEET

Parallel and perpendicular lines

Results! Why I have gotten a lot of results …



Thomas Edison

Rules

Parallel lines all have the same gradient.

$m_1 = m_2$

Perpendicular lines have gradients whose product is -1.

$m_1 \times m_2 = -1$

Solutions

Question	Parallel	Perpendicular	Neither
1	F	L	T
2	V	X	B
3	C	H	G
4	S	J	E
5	K	A	D
6	L	P	R
7	B	I	N
8	E	Y	F
9	D	M	P
10	H	Z	N
11	G	Q	U
12	O	H	K

Solutions

K 10

O 12

T $\frac{1}{2}$

W -6

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Questions

Determine whether the following pairs of lines (on the next page) are parallel, perpendicular or neither. The code letter for the puzzle appears in the correct column of the solutions in each case.

1 $y - 2x - 3 = 0$ and $2y + x + 2 = 0$

2 $y + 5x - 1 = 0$ and $3y + 15x + 7 = 0$

3 $2y + 3x - 4 = 0$ and $5y - 6x + 4 = 0$

4 $y + 2x - 3 = 0$ and $2y + 4x - 1 = 0$

5 $y + x - 3 = 0$ and $y - x + 4 = 0$

6 $3y - 8x + 2 = 0$ and $9y + 4x - 3 = 0$

7 $2y + 8x - 1 = 0$ and $4y - x + 8 = 0$

8 $y - 3x + \frac{1}{4} = 0$ and $2y - 6x + 7 = 0$

9 $\frac{1}{2}y + x - 1 = 0$ and $5y + 10x + 7 = 0$

10 $3y - 5x + 2 = 0$ and $2y + 7x + \frac{1}{4} = 0$

11 $y + 4x - 1 = 0$ and $2y - 8x + 3 = 0$

12 $2y + 3x + 7 = 0$ and $3y - 2x + \frac{1}{7} = 0$

13 If $2y + x - \frac{1}{2} = 0$ is parallel to $y + ax + 3 = 0$, what is the value of a?

14 If $5y - 2x + 6 = 0$ is perpendicular to $4y + bx - 7 = 0$, what is the value of b?

15 If $\frac{1}{2}y + 3x - 5 = 0$ is parallel to $2y + cx - 1 = 0$, what is the value of c?

16 If $2y + \frac{1}{3}x - 5 = 0$ is perpendicular to $y + dx + \frac{1}{4} = 0$, what is the value of d?

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